



Society of Petroleum Engineers

SPE-229091-MS

Unlocking Fuel Gas Reduction to Enhance Sustainability in Oil and Gas Operations

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229091-MS](https://doi.org/10.2118/229091-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Methane emissions pose a critical challenge in the oil and gas industry due to their high global warming potential. The purpose of this paper is to present a comprehensive study to reduce fuel gas consumption and methane emissions. In alignment with the company's sustainability agenda and methane mitigation strategy, a methodical approach has been implemented to reduce fuel gas consumption, which consists of 90% methane. This initiative encompasses modifications in operational philosophy augmented by dynamic process simulation, along with continuous monitoring and control mechanisms to ensure optimal fuel gas usage. This paper presents an 18% decline in fuel gas consumption, equating to 1002 MMSCF reduction at one of largest onshore gas oil separation facilities.

The Gas Oil Separation Plant (GOSP) processes crude oil extracted from over 600 wells, conducting phase separation into associated gas, crude oil, and produced water. The associated gas undergoes further processing, while dry lean gas (comprising approximately 90% methane) is recycled back into the GOSP plants and used in gas injection operations.

A dedicated task force was established to optimize the consumption of fuel gas, leveraging in-house expertise and implementing minimal procedural modifications. Utilizing a dynamic process simulation model, the team rapidly implemented several initiatives to achieve enhanced process optimization outcomes.

Post implementing this novel approach 18% reduction in fuel gas consumption was achieved and the results were tested over a period of six months. The benefits achieved for this initiative include:

- Optimization of gas injection compressor (gas turbine drive) during the hot season and running it in the winter, an automatic logic was developed to optimize and alert the DCS operator. This resulted in a net fuel gas saving of 13.5 MMSCF per annum.

- The substitution of nitrogen for fuel gas in three wash water tanks resulted in a savings of 230 MMSCF of fuel gas.

- The adjustment of the standby flare's logic achieved an annual reduction of 4.52 MMSCF.

- The optimization process applied to one of the four produced water tanks that yielded an annual savings of 394 MMSCF.

The modification of the oil flow suction tanks blanketing system led to a conservation of 360 MMSCF of fuel gas.

Implementation of smart Leak Detection and Repair (LDAR) program.

The dynamic optimization of fuel gas using process simulation software within the blanketing system, motive gas, flare pilots, and gas turbine fuel has resulted in notable benefits. These include a reduction in fugitive emissions, significant fuel gas savings, and enhanced operational efficiency. This optimization strategy can be implemented in other gas and oil separation plants and facilities.

Introduction

The Central Degassing Station (CDS) is a critical facility designed to manage and condition crude feedstock received from an extensive network of over 500 production wells. The incoming stream, which is typically high in water content and contains entrained gases, is routed through five parallel Gas Oil Separation Plant (GOSP) trains for initial bulk phase separation and stabilization. Each GOSP train is engineered to accommodate wet crude containing as much as 30% water by volume, ensuring robust operational flexibility under variable field conditions.

Within each train, primary separation is achieved via two sophisticated production separators arranged in series: the first-stage separator operates at high pressure (HP), efficiently separating free gas from the crude oil-water mixture while maintaining optimal upstream pressure for liquid handling. Downstream of the HP unit, the partially degassed emulsion is transferred to a second-stage intermediate-pressure (IP) three-phase separator. This vessel further enhances separation performance by removing residual dissolved gases and partitioning the fluid into crude oil, produced water, and remaining gas streams using advanced internals such as coalescing elements and weirs.

Trains 1 through 3 are equipped with integrated two-stage desalting and dehydration systems. These systems employ electrostatic treaters and chemical injection to reduce basic sediment and water (BS&W) levels and remove soluble salts from the crude, thereby meeting downstream export or refining specifications. Electrostatic desalters use an applied electric field to promote agglomeration and separation of fine water droplets from the oil phase, enhancing dehydration efficiency.

Conversely, Trains 4 and 5 utilize a downstream processing configuration in which oil from the second-stage separator is directed to surge vessels that provide residence time for gravity settling before entering dedicated dehydrators. These dehydrators, often fitted with heating and electrostatic grids, serve to break emulsions and ensure thorough removal of water from the crude stream. This staged approach allows for process optimization based on fluctuating inlet conditions, maintenance requirements, and operational reliability.

Throughout the CDS, instrumentation such as level controllers, pressure safety valves, and corrosion monitoring systems ensure safe, continuous operation, while integrated pipelines facilitate the routing of treated oil, separated water, and recovered gases to respective storage, reinjection, or gas processing facilities. This complex arrangement not only maximizes hydrocarbon recovery and product quality but also supports environmental compliance by minimizing flaring and effluent discharge.

The primary purpose of processing crude oil is to separate the associated water, which can be classified as either free water or emulsified water. Free water typically separates through gravity-based methods within the initial and secondary stage separators. Emulsified water is removed by adding chemicals and fresh water; demulsifiers are commonly used to break the emulsion. Electrostatic coalescence may also be employed to improve water separation.

After separating water from the crude oil, the treated product flows through a shared header to degassing boots, with three identical units in operation. To maintain export crude within Reid Vapor Pressure (RVP) specifications, a stripping stream containing lean gas is injected into the inlet line of each degassing boot, facilitating the removal of lighter components before the crude enters the low-pressure (LP) header. Each

degassing boot can operate as a gas/liquid separator at oil production capacities equivalent to two production trains.

Crude oil from the degassing boots is transferred to a Flow Suction Tank (FST), where further water/sediment settling and gas separation occur. The FSTs act as buffer tanks to store and pump export-grade crude. Outlets from all FSTs merge into a single header supplying five booster pumps working in parallel. These pumps raise the pressure of the export crude before it moves to three Main Oil Line (MOL) pumps arranged in series, which provide additional pressure for distribution via two spur lines to the primary oil network.

Gas streams from the initial separators (high-pressure gas), secondary separators (intermediate-pressure gas), surge vessels, degassing boots, and tanks (low-pressure gas) are directed to the Gas Processing facilities through three sets of gas headers based on pressure levels. High-pressure gas is routed to the Gas Lift Compression Station (GLCS).

Produced water from production and test separators, the Dehydration System, and flow suction tanks is collected in four disposal water tanks—two new and two old—before transfer to disposal water wells and produced water reinjection manifolds via a pumping system. Each tank is equipped with an oil skimmer to maintain oil-in-water specifications. All disposal water tanks operate in parallel, except during maintenance when individual tanks may be shut down.

Reservoir pressure is managed by a gas injection plant using two compressors: one powered by a fuel gas-driven turbine and the other by an electric motor. Compressor operation is determined by gas injection requirements, with power supplied by gas turbines acting as backup for the grid power station.

Lean gas (90% methane) is utilized at the central degassing station as blanketing, pilot, purge, and stripping gas, as well as motive gas in ejectors.

Field Background and Challenges

The oil and gas separation process operates at an industrial scale, employing a range of processing vessels, separators, and compression equipment.

A multidisciplinary task force was established to conduct a comprehensive assessment of fuel gas consumers within the facility and to identify technically viable solutions for mitigating methane emissions and optimizing fuel gas utilization in compliance with safety and regulatory requirements.

- Comprehensive mapping and classification of all fuel gas end-users, including the Central Degassing Station (CDS), Injection Plant, and the onsite Power Generation Unit.
- Systematic review of current operational protocols to pinpoint inefficiencies and potential areas for process optimization.
- Deployment of a Leak Detection and Repair (LDAR) program aimed at quantifying and reducing fugitive emissions throughout the gas system.
- Identification and prioritization of immediate mitigation measures, alongside the formulation of both tactical (short-term) and strategic (long-term) action frameworks.

Lean gas system

Lean gas is delivered to the CDS Plant from an external supplier that processes associated gas and provides a portion as lean dry recycle gas. As indicated in the schematic below, the primary 12-inch header distributes lean gas for the following operational purposes:

- Stripping gas in the degassing boot to achieve crude oil Reid Vapor Pressure (RVP) compliance
- Blanketing gas for fresh water tanks, disposal water tanks, and flow suction tanks to prevent ingress of air and maintain inert atmospheres

- Motive gas supply for tank ejector operation
- Pilot and purge gas streams serving high-pressure (HP), low-pressure (LP), and dedicated flow suction tank flare systems

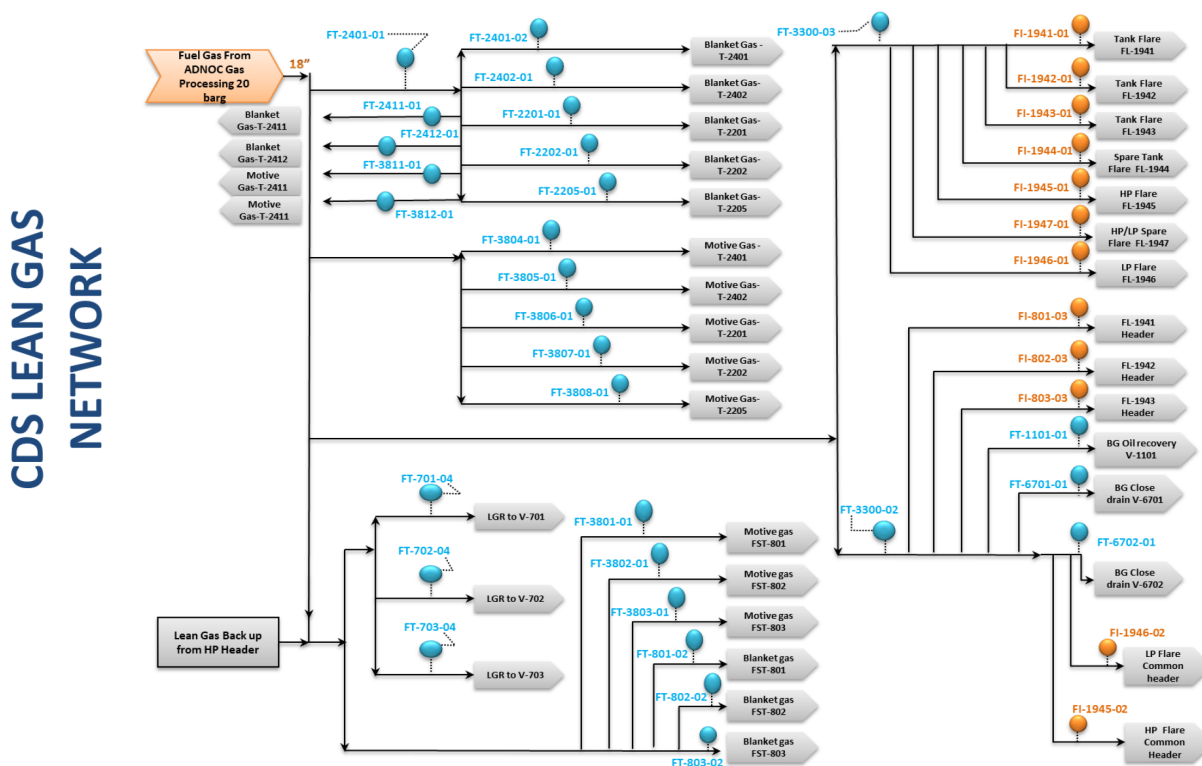


Figure 1—Lean gas network of CDS

Quick implementation

This category comprises initiatives that necessitate no capital expenditure, involve only minimal alterations to current operating protocols or organizational philosophy, and require updates to control frameworks. These initiatives are designed for rapid deployment, including:

Operational Optimization Measures: Structured to enhance legacy workflows and system efficiencies without large-scale reengineering.

Expedited Implementation Protocols: Utilizing internal functional capabilities for swift execution and integration within existing operational structures.

Long term implementation

Such modifications are characterized by high complexity, necessitating a multi-year (typically exceeding three years) implementation timeline. They often entail substantial organizational re-engineering or the integration of advanced technological solutions.

Compressors

The Train-1 compressor is integrated with a gas turbine driver, whereas Train-2 employs an electric motor for gas injection operations. Operational scheduling entails Train-1 functioning for one month per quarter, with Train-2 allocated two months per quarter, to reduce turbine exhaust venting emissions.

Simulation Basis and Assumptions:

- Simulation Software: Aspen HYSYS V11.2.1
- Thermodynamic Model: Peng-Robinson Equation of State applied for all fluid package calculations, implemented with the following configuration options:

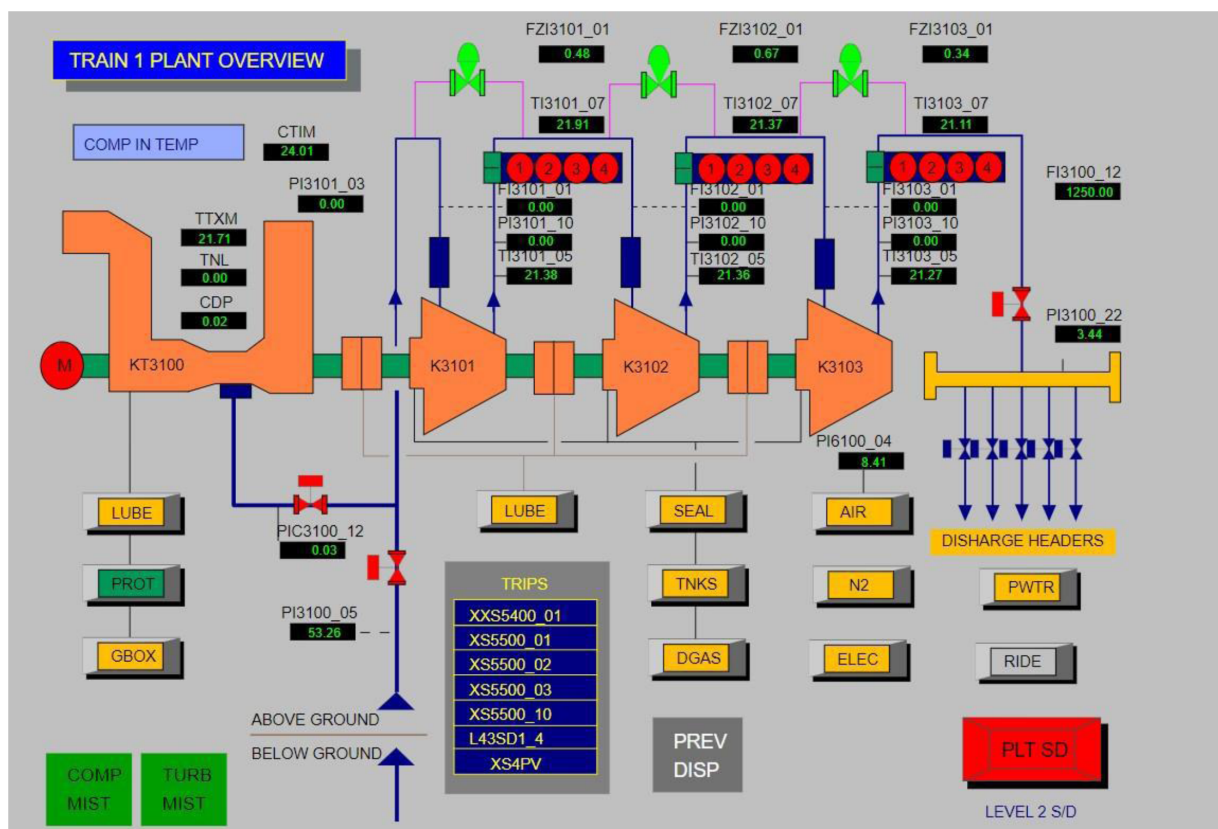


Figure 2—Gas injection train 1

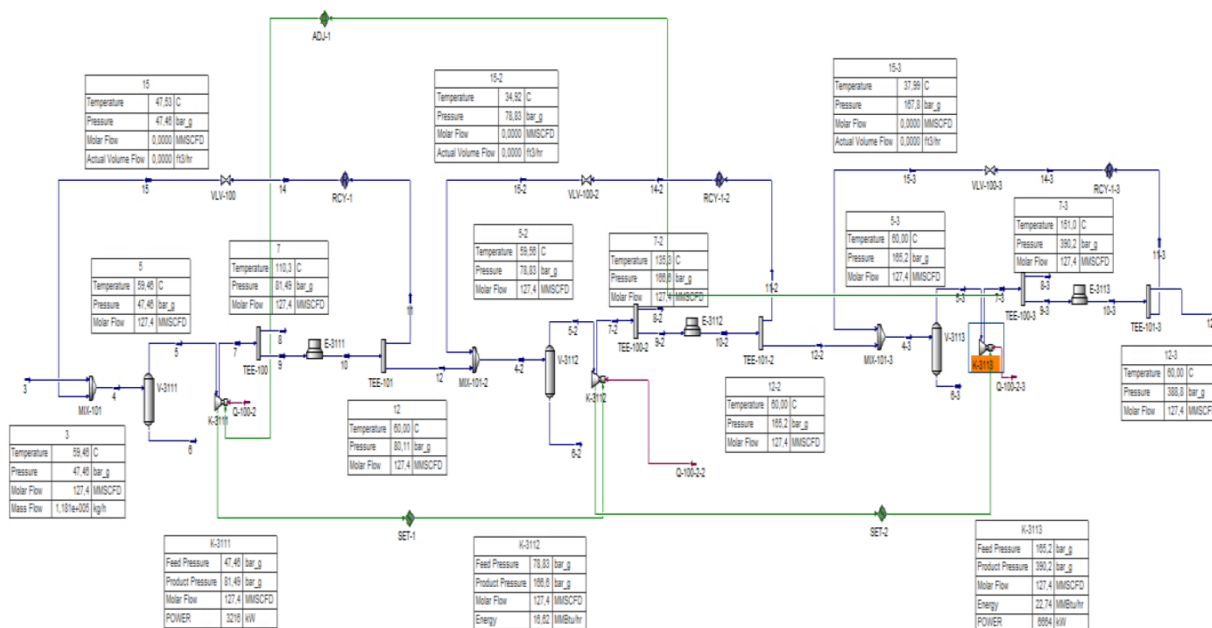


Figure 3—Dynamic simulation of gas injection compressor train

The gas injection compressor for Train 1 is driven by a KT-3100 gas turbine, which is operated for one month per quarter while Train 2, equipped with an electric drive, operates for the remaining two months to optimize exhaust venting protocols. The KT-3100 gas turbine receives fuel gas from the K-3101 compressor suction drum, with flow measurement compensated and monitored via flow transmitter FT-3100-13. Annually, operational scheduling allocates approximately one-third of the duty cycle to Train 1 and two-thirds to Train 2, with Train 1 scheduled primarily during the summer period. Due to reduced thermodynamic efficiency of gas turbines in elevated ambient temperatures, it is preferable—based on performance analysis—to prioritize Train 1 operation during lower temperature periods and reserve Train 2 (electric drive) for higher temperature seasons. A case study recommends allocating four months of winter duty to Train 1 and eight months of summer duty to Train 2. Table provides detailed data regarding Train 1's fuel gas consumption under winter operating conditions.

Month			Power Index	KT3100	Mech. Power	Modified mech. Power	Modified
temp deg F		temp deg C		gas MMSCF		KT3100 gas	
Jan	67	19.44	0.267	73.6	19.66	19.66	73.60
Feb	69	20.56	0.266	4.7	1.25	1.25	4.70
Mar	73	22.78	0.263	7.6	2.00	42.91	163.29
Apr	80	26.67	0.258	166.5	42.91	2.00	7.75
May	88	31.11	0.252	3.9	0.98	0.98	3.90
Jun	91	32.78	0.250	2.3	0.57	0.57	2.28
Jul	94	34.44	0.248	172.6	42.74	3.57	14.42
Aug	94	34.44	0.248	6.6	1.63	1.63	6.58
Sep	91	32.78	0.250	0	0.00	0.00	0.00
Oct	85	29.44	0.254	168.8	42.90	42.74	168.18
Nov	78	25.56	0.259	3.8	0.98	0.99	3.80
Dec	71	21.67	0.264	13.5	3.57	42.90	162.34
623.9		159.20		159.19		610.85	
MMSCF Gas		623.9			610.85		

This solution can be executed with zero capital expenditure. An optimized automation routine is proposed within the Distributed Control System (DCS), facilitating real-time computation of operational runtimes for Train 1 and Train 2, and generating automatic operator alerts to prompt train changeovers as necessary.

Implementation of the updated operational protocol for the Train 1 Gas Injection Compressor (GIC) subsystem yields an annualized reduction in fuel gas consumption by 13.05 million standard cubic feet (MMSCF), corresponding to projected yearly cost savings of \$7,800.

Flare system

The CDS flare assembly incorporates a High Energy Ignition System (HEIS), with redundancy provided by a Front Flame Generator (FFG). Continuous pilot functionality is ensured via uninterrupted fuel gas delivery. Standard configuration assigns one flare as standby, facilitating backup operations in the event of HP or LP flare malfunction.

Following task force directives, modifications to operational protocols led to the isolation of pilot fuel gas on the standby flare system. Standby flare ignition integrity is verified through routine operator assessments, while purging is executed utilizing nitrogen injection into the flare header. The updated flare transition procedure has optimized fuel gas consumption associated with the standby flare.

Analogous methodologies were implemented for the gas injection compressor train standby flare. Modifications at the gas letdown station now permit interconnection with both trains' flares, effectively ceasing fuel gas allocation to the standby system.

Operational enhancements resulted in annual fuel gas conservation of 4.5 MMSCF and a reduction of 321 metric tons in CO₂ emissions.

Fresh water Tanks

The central degassing station utilizes three storage tanks and employs fuel gas as both the blanketing medium and the motive force for process ejectors. Presently, wash water injection rates are maintained at 16,000–18,000 BWPD, which is substantially lower than the original design specification of 30,000 BWPD. Operational protocol adjustments have been implemented, including the isolation of one potable water storage tank maintained under a regulated positive nitrogen blanket for emergency deployment. This procedural modification has resulted in an annual reduction in fuel gas consumption equating to approximately 76.65 MMSCF.

Strategically, the facility aims to transition the fresh water blanketing system from fuel gas to nitrogen in the long term. Fresh water supply is sourced from ADEWA, ensuring the absence of hydrocarbons and compliance with treatment standards. A Management of Change (MOC) request has been initiated to conduct a feasibility study on the permanent conversion to nitrogen blanketing and to determine the capacity adequacy of the existing nitrogen skid infrastructure. Anticipated outcomes include a reduction in recycled gas allocations to affiliate entities, improved process energy efficiency, decreased operational expenditure associated with fuel usage, and mitigation of methane emissions, encompassing both point source and fugitive sources.

Projected annualized savings consequent to full nitrogen system implementation are estimated at 230 MMSCF of fuel gas.

Disposal Water Tanks

The Central Disposal System (CDS) comprises four produced water disposal tanks, each with a nominal capacity of 100,000 barrels of water per day (BWPD). Fuel gas serves both as the blanket medium and as motive force within the ejector system. Current operational requirements are met with two tanks; therefore, the operating procedure has been modified to maintain one tank under continuous positive nitrogen blanketing. This revision results in an annual reduction of fuel gas consumption by approximately 394 million standard cubic feet (MMSCF).

HP gas scrubber

The central degassing station currently uses HP associated gas as a backup for the lean gas system. The potential use of this setup as a permanent fuel gas supply is being evaluated, with an anticipated savings of 20 MMSCFD of fuel gas per day.

Lean Gas recycle to degassing boots

Degassing boots in the Crude Distillation System (CDS) are utilized to control the Reid Vapor Pressure (RVP) of export-grade crude oil by employing lean gas—primarily methane—as the stripping medium. A controlled summer operational trial demonstrated a reduction in lean gas injection rates from 9 MMSCFD to 1.5 MMSCFD, with no observed deviation in RVP compliance, thus achieving a net fuel gas conservation of 8 MMSCFD over the test period.

Leak Detection and Repair LDAR

Within a robust methane emission quantification and abatement protocol addressing fugitive, venting, and incomplete combustion sources, an advanced Leak Detection and Repair (LDAR) initiative was executed at across all legacy assets. The 2024 surveillance campaign entailed systematic inspection of potential leak points and direct measurement of methane flux from discretely identified emission sites. Comprehensive integrity assessments were conducted on process equipment, isolation devices, and flange assemblies to localize point-source methane emissions, informing a component-level rectification regime aimed at optimizing fugitive loss abatement.

The monitoring framework deployed technologies such as Toxic Vapor Analyzer (TVA), Optical Gas Imaging (OGI) using infrared spectroscopy, and Heated Flame Ionizations Detectors (HFS) for high-sensitivity detection and quantification of hydrocarbon releases.

Fugitive Emission Sources

Piping Interfaces and Mechanical Joints

- Flange gaskets
- Threaded union
- Tubing couplers and miscellaneous sealed interfaces

Isolation & Actuation Valves

- Gate valve stem packing
- Ball valve stem packing
- Control valve bonnet/packing
- Plug valve packing and additional gland configurations

Pressure Relief Assemblies

- Pressure safety valve seats/outlets
- Rupture disc assemblies and allied pressure relief apparatus

Atmospheric Process Openings

- Unsealed process vent stacks
- Miscellaneous atmospheric vent points not classified within OGMP defined venting streams

Dynamic Equipment

- Pump mechanical seals
- Compressor shaft and packing seals

Venting Emission Sources:

- Centrifugal compressor labyrinth/seal leakage
- Reciprocating compressor rod packings
- TEG glycol dehydration systems
- Atmospheric storage tanks and vessels
- Plunger lift and wellbore liquids unloading events
- Annular/casing head venting
- Hydraulic fracturing flowback/completions
- Process vents
- Auxiliary sources

Remediation strategies for each identified emission source were categorized according to operational continuity constraints. Immediate corrective maintenance was performed where system shutdowns were non-essential, while interventions requiring plant outage were integrated into scheduled turnaround frameworks and executed accordingly. Post-LDAR implementation analytics confirmed a net reduction in fugitive methane emissions by 4,000 metric tons.

Conclusion

In summary, methane emissions are a critical factor in the oil and gas sector owing to their high global warming potential. To align with the organization's sustainability objectives and methane abatement strategies, a systematic methodology was implemented for fuel gas optimization—primarily targeting methane-rich streams. This methodology encompassed revisions to standard operating procedures underpinned by dynamic process simulation models, alongside continuous monitoring and supervisory control systems for real-time management of fuel gas consumption. The study documents an 18% decrease in fuel gas utilization, corresponding to 1002 MMSCF equivalent to 5492 metric tons of Co₂, achieved at one of largest onshore Gas Oil Separation Plants (GOSPs). This optimization framework can be scaled for use at other gas and oil separation sites.

Acknowledgement

The authors would like to thank Company Management for their approval and support in publishing and present this paper, which will benefit others operating gas oil separation plants. The authors also appreciate the efforts and collaboration of the Operations and Maintenance teams of the Asset, who played a vital role in the successful implementation of this work.

Glossary

- CDS: Central degassing station
- BWPD: Barrels of water per day
- GOSP: Gas oil separation plant
- HP: High pressure
- LP: Low pressure

IP: Intermediate pressure
MOL: Main oil line
FST: Flow suction tank
RVP: Reid vapor pressure
PT: Pressure transmitter
DWT: Disposal water tank
FWT: Fresh water tank
GIP: Gas injection plant

References

CDS operating manual, Gas injection plant operation manual, Leak detection and repair report.
SPE-220443-MS - Let's talk flare – Centralized Flare Control and Management for Upstream Facilities. *Flaring diagnostic tool by deploying Mathematical Model*